

ORIGINAL ARTICLE

Mixed species biofilms of *Lactobacillus plantarum* and *Listeria innocua* show facilitated entrance to the VBNC state during chlorine-induced stress

Magdalena A. Olszewska¹  | Ireneusz Białobrzewski²

¹Department of Industrial and Food Microbiology, Faculty of Food Science, University of Warmia and Mazury in Olsztyn, Olsztyn, Poland

²Department of Systems Engineering, Faculty of Technical Sciences, University of Warmia and Mazury in Olsztyn, Olsztyn, Poland

Correspondence

Magdalena A. Olszewska, Department of Industrial and Food Microbiology, Faculty of Food Science, University of Warmia and Mazury in Olsztyn, Plac Cieszyński 1, 10-726 Olsztyn, Poland.
Email: magdalena.olszewska@uwm.edu.pl

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Abstract

Biofilms of *Lactobacillus plantarum* and *Listeria innocua* were grown under single- and mixed-culture conditions in a bioreactor with either half-strength brain heart infusion (BHI) medium or BHI medium with sodium dichloroisocyanurate (150 ppm) for 24 hr. A quantitative approach using plate counts and fluorescence in situ hybridization (FISH) was applied, representing the culturable and viable cells, respectively. Species contribution to the single and mixed species biofilms was similar, reaching ca. 6.5 and 7.0 log cfu (cells)/mL for *L. plantarum* and *L. innocua*, respectively. In the presence of chlorine, species contribution to the single or mixed species biofilms was 10- to 100-times lower than in the absence of this chemical for *L. plantarum* and *L. innocua*, respectively. Notably, a large difference was observed between the culturable and the viable cells in the mixed species biofilms, reaching 1.5 and 2.5 log units for *L. plantarum* and *L. innocua*, respectively. *L. innocua* achieved almost complete nonculturability during an early stage of biofilm growth in the coculture, indicating that the physiological state of cells was significantly affected not only by chlorine challenge but also by inter-species interactions. Importantly, all biofilms consisted of culturable cells during the late stage of growth, which represented a potential source of contamination.

Practical Applications

Fluorescence-based methods for bacterial viability measurement have remained useful for a wide diversity of applications. In our study, the approach employing FISH in a parallel with plate counting allowed us to enumerate viable and culturable cells of *L. plantarum* and *L. innocua* under either single- or mixed-culture conditions. Thus, we recommend it for accurate estimation of viable but nonculturable cells in the experimental set-up comprising single-/mixed-culture sessile community.

1 | INTRODUCTION

Lactobacillus plantarum is known as spoilage contaminant of food products, with biofilms found in food processing environments representing a potential source of contamination (Kubota, Senda, Nomura, Tokuda, & Uchiyama, 2008). Biofilms consist of microorganisms attached to a surface embedded in an extracellular polymeric substance matrix

(Costerton, Stewart, & Greenberg, 1999). Most studies on *L. plantarum* biofilm formation focus either on the variation between strains in their capability to form biofilms in different conditions (Fernández Ramírez, Kostopoulos, Smid, Nierop Groot, & Abee, 2017; Kubota et al., 2008) or on the resistance of these biofilms to chemical treatments (Kubota et al., 2008; Kubota, Senda, Tokuda, Uchiyama, & Nomura, 2009). However, in a variety of environments, including food processing areas,

L. plantarum will most likely grow on surfaces with other microorganisms in a mixed species biofilm. Previous research by Van der Veen and Abee (2011) exemplifying this, as it shows *L. plantarum* to be able to form biofilms with *Listeria monocytogenes*—representing a higher food spoilage potential and a safety risk—under static conditions. Moreover, multiple species of *Listeria*, including *L. monocytogenes*, is found in similar niches as *L. plantarum*, and both have been previously isolated from such food products as green table olives (Caggia, Randazzo, Di Salvo, Romeo, & Giudici, 2004) or fermented sausages (Tolvanen et al., 2008), which suggests that these microbes can stably coexist. Hence, the aim of this study was to assess the population dynamics of *L. plantarum* and *Listeria innocua* in single and mixed species cultures in a biofilm reactor under dynamic conditions using two methods of enumeration: plate count method and fluorescence in situ hybridization (FISH) with species-specific rRNA probes.

Most reports comparing biofilm formation by *L. plantarum* are based on the crystal violet (CV) assay, which ensures a global quantification and not a detailed characterization of biofilms including the physiological state of cells and changes in cell morphology during growth under particular conditions. FISH has previously been employed to monitor population dynamics of LAB during cheese ripening (Mikš-Krajnik & Babuchowski, 2014) and for identification of LAB in oral biofilms (Zijngel et al., 2010), thus proving useful in analyses of samples from complex matrices. The utilization of FISH allows the enumeration of cells in the biofilm with rRNA content high enough to detect the viable cell part that following dispersal, evaluates the biofilm outgrowth potential. Furthermore, in mixed species biofilms, this assay can be developed to detect the viability of specific bacteria using appropriate probes (Joux & Lebaron, 2000). In turn, the plate count technique enables the estimation of culturable cells so that by comparing the number of viable cells to that of culturable cells in the sample the abundance of viable but non-culturable (VBNC) cells can be determined (Li, Mendis, Trigui, Oliver, & Faucher, 2014). Therefore, this study contributes to a better understanding of *L. plantarum* and *Listeria* at the level of viable and culturable cells and provides a model of dynamic biofilm growth in conditions simulating natural settings including food processing environments. The food isolates, *L. plantarum* B1 and *Listeria innocua* FI2, which showed the highest biofilm formation under static conditions in a prescreen using a CV assay (data not shown), were selected for this study. Because of the associated biosafety risk, we examined *L. innocua* that is genetically, ecologically, and phenotypically close to *L. monocytogenes* (Bonifácio et al., 2018).

2 | MATERIALS AND METHODS

2.1 | Bacteria and media

Food isolates, *L. plantarum* B1 and *L. innocua* FI2, stored in the collection of the Chair of Industrial and Food Microbiology, University of Warmia and Mazury in Olsztyn, were streaked from -20°C stocks in De Man, Rogosa and Sharpe agar (MRS-agar; Merck, Darmstadt, Germany) and Tryptic soy agar (TSA, Becton Dickinson, Sparks, MD), and incubated for 48 hr anaerobically at 30°C and aerobically at 37°C , respectively. Single colonies of B1 and FI2 were inoculated in 10 mL

of MRS and brain heart infusion (BHI; Becton Dickinson) broths for 18 hr to prepare the running cultures. Biofilm growth was performed in BHI with the addition of 0.005% manganese sulfate (BHI-Mn; Merck) which was previously shown to increase the contribution of *L. plantarum* (Van der Veen & Abee, 2011).

2.2 | Biofilm growth

Biofilms were grown in the Center for Disease Control and Prevention's CDC Biofilm Reactor[®] (BioSurface Technologies Corporation, Bozeman, MT). The CDC biofilm reactor[®] contains eight rods, each of which holds three polycarbonate (PC) coupons (12.7 mm diameter). An overnight culture was prepared by incubating B1 or FI2 in BHI-Mn for 18 hr at 30°C or 37°C , respectively. A sterile reactor was inoculated with 3.5 mL of an overnight culture that had been added to 350 mL of half-strength BHI-Mn. The initial B1, FI2 or B1/FI2 population in the batch medium was ca. 10^6 cfu per milliliter. The reactor was first operated for 1 hr at 70 rpm at 30°C (the attachment step) and subsequently it was connected via a nutrient feed line to a carboy containing the medium with or without a sodium dichloroisocyanurate solution (150 ppm; Ecolab, Krakow, Poland) (the biofilm formation step) and was operated in the continuous-flow mode at a flow rate of 4 mL/min for 24 hr. At each sampling time point, a rod was removed from the reactor and the coupons were collected in Falcon tubes containing 0.1 M phosphate buffer saline (PBS, Sigma, St. Louis, MO). The tubes were sonicated for 1 min and vortexed for 2 min to remove the biofilms to be used for plate counting and FISH.

2.3 | Analyses

Biofilm samples were serially diluted in saline and spread-plated on MRS-agar and Oxford-agar (Becton Dickinson) plates. The characteristic lactobacilli and listeriae colonies grown on the plates were counted after prolonged to 72 hr of incubation at 30°C under anaerobic conditions (Anaerocult[®] C; Merck) and at 37°C under aerobic conditions, respectively. Prior to the hybridization step, biofilm samples were filtered onto black PC filters (13 mm; pore size 0.22 μm ; Isopore[®], Merck). Cells were fixed by overlaying the filters twice with 3 mL of PBS; after removing this solution, 3 mL of 96% ethanol was added, allowed to act for 3 min at room temperature and then removed. After fixation, the cells were permeabilized with 200 μL of a 10 mg/mL lysozyme solution (in 10 mM Tris, 1 mM EDTA; Merck); after 5 min the solution was removed by applying vacuum. An additional washing with distilled water was done to remove lysozyme remains. Air-dried filters were stored at -20°C until hybridization. Selected 16S rRNA probes fluorescently labeled at $-5'$ end with Cy3 and PB were purchased from Bionovo (Legnica, Poland) and are listed in Table 1. The filters were placed on glass slides and covered with 20 μL of the hybridization solution, composed of 18 μL of a hybridization buffer (20 mM Tris-HCl pH 7.2, Sigma-Aldrich, Poznan, Poland; 0.9 M NaCl, Stanlab, Lublin, Poland; 0.01% sodium dodecyl sulfate, SDS, Sigma-Aldrich) and 2 μL of a 50 ng/ μL fluorescent oligonucleotide probe solution. Then, they were transferred into a hybridization chamber, equilibrated with the

TABLE 1 The characteristics of the probes used in this study

Probe	5'-3' sequence	Target	Labeling	Specificity	Reference
LbpV3	CCG TCA ATA CCT GAA CAG	16S	Cy3	<i>L. plantarum</i>	Ercolini, Hill, and Dodd (2003)
LisUn-3	CCC CAA CTT ACA GGC	16S	PB	<i>Listeria</i> spp.	Brehm-Stecher, Hyldig-Nielsen, and Johnson (2005)

Abbreviations: Cy3, cyanine 3 dyes; PB, Pacific blue.

hybridization buffer as described above and incubated at 40°C for 3 hr (GFL Incubator, Burgwedel, Germany). After the hybridization step, the filters were transferred to Falcon tubes containing a prewarmed (to 48°C) washing solution (20 mM Tris-HCl pH 7.2, 0.9 M NaCl, 0.01% SDS) and freely incubated floating at 48°C for 20 min (GFL Incubator). After the washing step, the filters were rinsed with ddH₂O for 1 min in the dark. Air-dried filters were mounted on glass slides with Citifluor AF1:VectaShield (4:1, vol/vol) antifading reactants (Citifluor Ltd, London, England and Linaris, Wertheim, Germany, respectively) and stored at -20°C until microscopic analysis. Biofilm samples were imaged with a Olympus BX51 epifluorescent microscope (Olympus, Hamburg, Germany). Images were analyzed using the CellSens Dimension System 1.5 (Olympus). Each experiment with single (B1/FI2) and mixed (B1 + FI2) species biofilms in chlorine/no chlorine conditions included three replicate coupons. Plate count analysis was conducted in triplicate, and more than 10 images per sample were taken for FISH analysis. The mean values were further calculated as log cfu/mL.

2.4 | Data analysis

To determine whether single and mixed species biofilms showed differences in the population dynamics during the simulation in the CDC Biofilm Reactor®, the growth and death curves were fitted with the McKellar model using two differential equations Equations (1) and (2) (McKellar & Lu, 2004). Model fit to experimental data was evaluated based on the coefficient of determination (R^2).

$$\frac{dX_d}{dt} = X_d \cdot \mu_{\max,d} \cdot \left(1 - \frac{X_d}{X_{\max,d}}\right) \quad (1)$$

$$\frac{dX_r}{dt} = X_r \cdot \mu_{\max,r} \cdot \left(1 - \left(\frac{X_r}{X_{\max,r}}\right)^{0.5}\right) \quad (2)$$

X	-	number of cells	cfu (cells)/mL
t	-	time	(h)
$\mu_{\max}$	-	maximum growth/death rate	(1/h)
$X_{\max}$	-	maximum cell density	cfu(cells)/mL

Lowercase indexes:

d, death; r, growth.

The model was implemented in the MATLAB environment (MathWorks) using an additional optimization procedure, knitromatlab (Artelys, France) to determine four coefficients (Table S1) in the model, whereas the initial values, X_d and X_r , were determined based on the trial and error method.

3 | RESULTS

The curves plotted for the single and mixed species biofilms of *L. plantarum* B1 and *L. innocua* FI2 grown dynamically without chlorine are shown in Figure 1. For the single species biofilm of *L. plantarum* B1, the plate count curve increased smoothly from the initial level of 5.5 log cfu/mL to 6.5 log cfu/mL and from the 8th hr reached a growth plateau (Figure 1a). An increase of 1.0 log unit was observed in the FISH count curve too and thus the plate and FISH curves almost overlapped during the period between 8 and 24 hr. As for *L. innocua* FI2, the plate count curve decreased sharply by 1.0 log cfu/mL within 2 hr followed by a successive growth up to ca. 7.0 log cfu/mL, while the FISH count curve decreased more smoothly (a 0.2-log-unit cell reduction) followed by a fairly constant growth observed in the period between 6 and 20 hr (Figure 1b).

In the mixed species biofilm, there was a small increase in FISH cell counts over the 16-hr period followed by a growth plateau at the level of ca. 6 log cells/mL for *L. plantarum* B1 (Figure 1c). In contrast, the plate counts showed a gradual decrease by a 0.5 log cfu/mL and from ca. 8th hr onward the plate count curve increased smoothly. As for *L. innocua* FI2, a substantial difference between FISH and plate counts was observed over the period of 13 hr (Figure 1d). Within 7 hr, FISH counts increased by ca. 1.0 log cells/mL, while plate counts decreased by 0.5 log cfu/mL and then during the period between 7 and 13 hr intensively increased up to ca. 7.0 log cfu/mL. The cell growth reached a plateau at 15th hr with no difference between plate and FISH counts.

The curves plotted for the single and mixed species biofilms grown with chlorine are shown in Figure 2. In the single species biofilm, the plate count curve of *L. plantarum* B1 declined sharply from an initial cell count of ca. 4.0 to 0.5 log cfu/mL within 3 hr (Figure 2a). Subsequently, a period between 3 and 7 hr was characterized by a rapid increase up to 5.0 log cfu/mL followed by a growth plateau. In contrast, FISH count curve decreased from ca. 6.5 to 6.0 log cells/mL within 3 hr followed by a slight increase and then a slow but steady decrease. For *L. plantarum* B1, there were no further changes in the physiological state of cells, as the FISH and plate count curves overlapped from ca. 20th hr onward. Also for *L. innocua* FI2, the difference between plate and FISH cell counts was greater in the presence of chlorine than at its absence, however the magnitudes were different (Figure 2b). We found that the plate counts decreased sharply, while the FISH counts decreased smoothly, reaching 4.0- and 0.5-log-unit cell reductions, respectively within 3 hr. Although the plate counts increased between the period of 8 and 20 hr, the plateau levels of

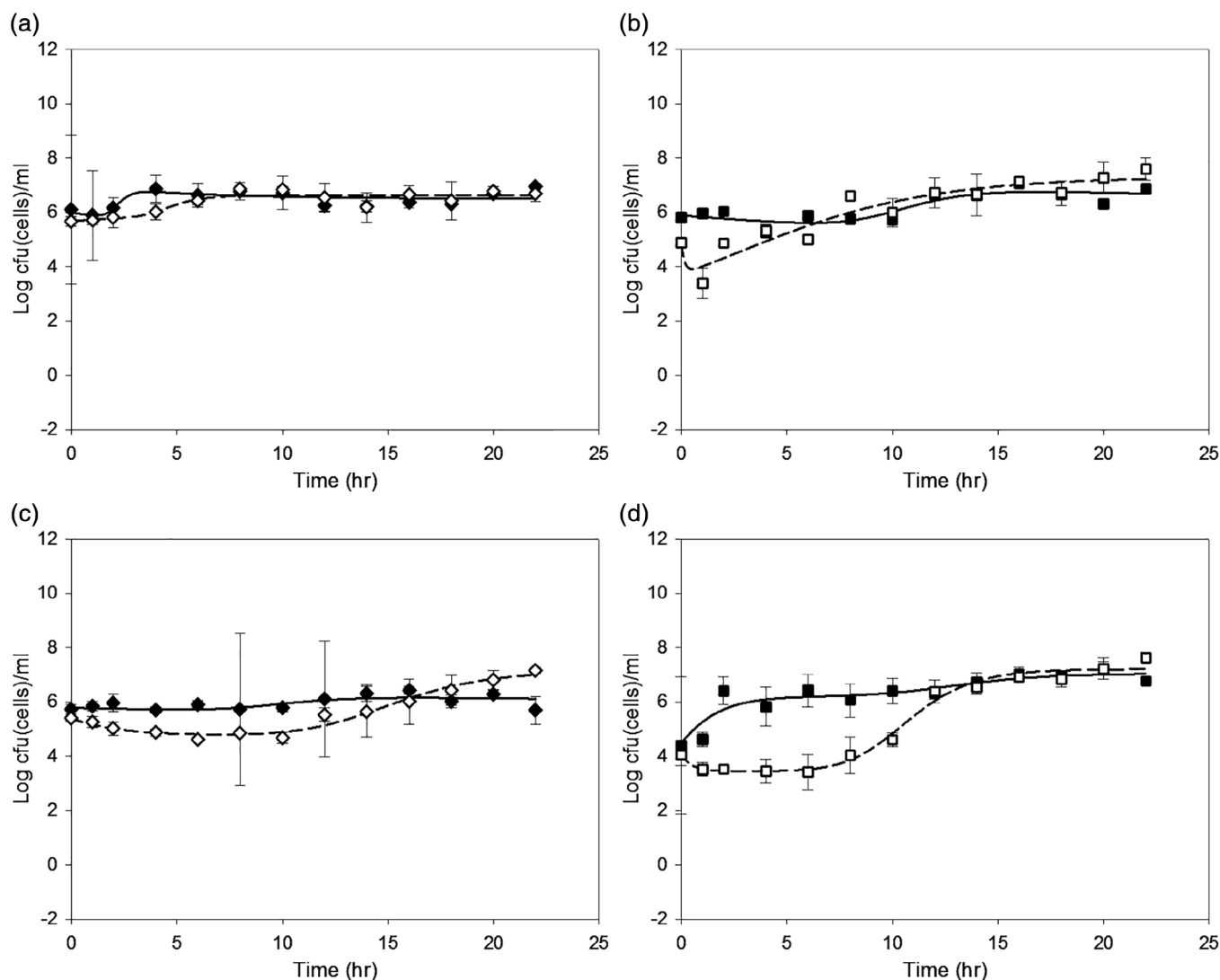


FIGURE 1 Single (a, b) and mixed (c, d) species biofilm growth in a CDC biofilm reactor® under conditions as described in Section 2 without the presence of chlorine. The graphs show the experimental data points (mean $\pm$ SD) and model curves of *L. plantarum* B1 (diamonds) and *L. innocua* FI2 (squares) determined by plate count method (open symbols and dotted line) and FISH (closed symbols and continued line). FISH, fluorescence in situ hybridization

ca. 4.0 and 5.5 log cfu (cells)/mL were reached for plate and FISH counts, respectively.

In the mixed species biofilm, the qualitative trends of FISH curves were similar to those of single species biofilms for both *L. plantarum* B1 and *L. innocua* FI2, reaching a final cell count of ca. 6.0 log cells/mL. Initial cell counts as low as 3.0 and 1.5 log cfu/mL were obtained with the plate count method for *L. plantarum* B1 and *L. innocua* FI2, respectively. As for *L. plantarum* B1, the plate count curve further decreased during an 8-hr period after which it started to increase, reaching the ca. 4.0 log cfu/mL plateau (Figure 2c). In contrast, the plate counts of *L. innocua* FI2 dropped below detection limit and then increased up to 3.5 log cfu/mL (Figure 2d).

L. plantarum B1 and *L. innocua* FI2 cells exhibited the characteristic rod-shaped morphology. As for *L. innocua* FI2 cells, their size varied from ca. 1.8 μ m to ca. 2.6 μ m (Figure 3a). As for *L. plantarum* B1 cells,

no substantial change was observed in cell morphology, regardless of conditions, whereas smaller forms of ca. 1.1 μ m were more frequently shown in the case of *L. innocua* FI2 cells. In turn, cell fading was a common phenomenon for *L. plantarum* B1 (Figure 3b–d).

4 | DISCUSSION

Even though there is much information available on biofilm formation by single species, it is likely that in a variety of environments including food processing areas, biofilms are formed by multiple bacterial species. To date, little work has been done on the performance of *L. plantarum* and *Listeria* cells in mixed species biofilms. A vast majority of studies have focused on the single species biofilms of either *L. plantarum* or *Listeria* and their resistance to disinfectants with the special emphasis on the *L. monocytogenes* due to the potential food

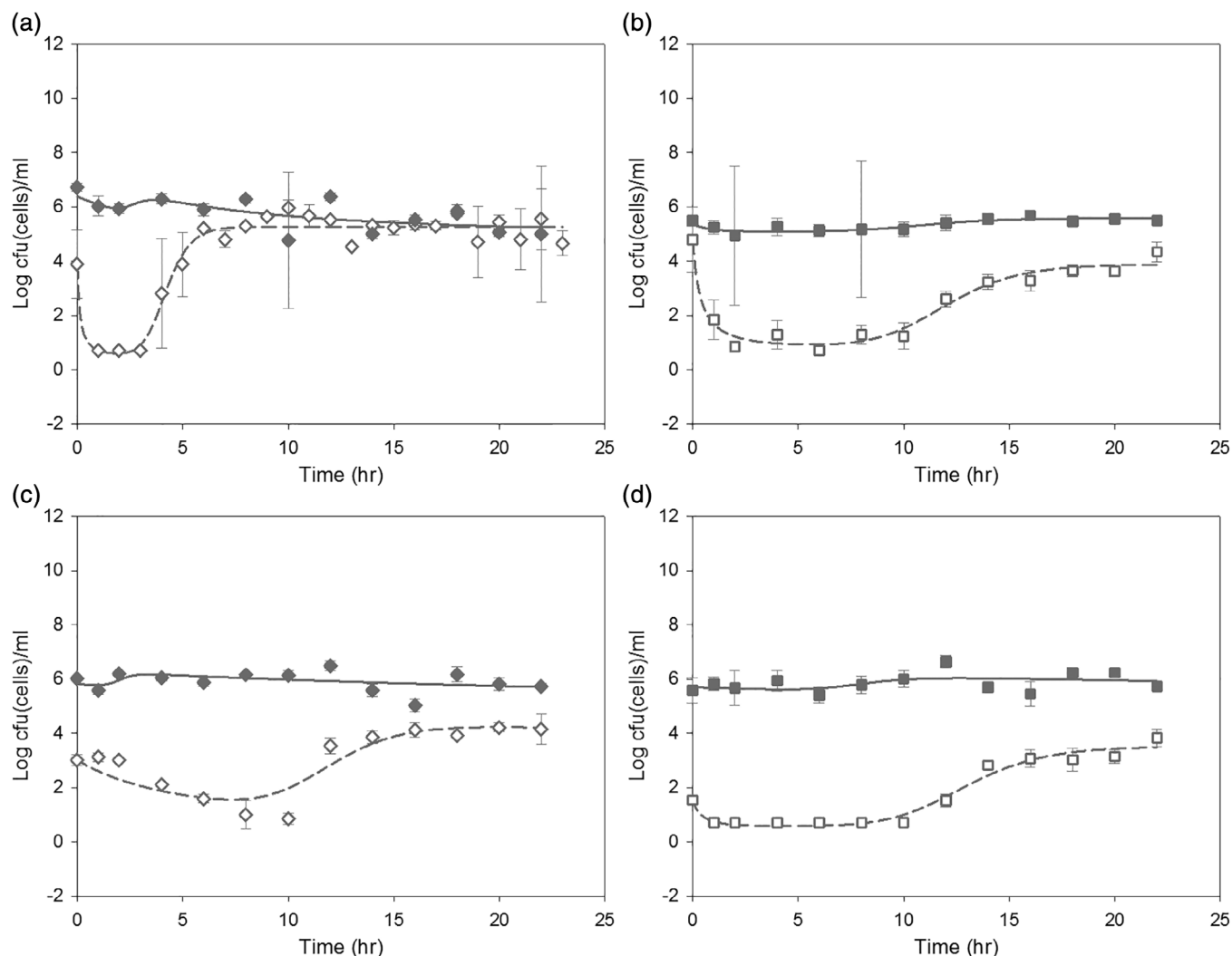


FIGURE 2 Single (a, b) and mixed (c, d) species biofilm growth in a CDC biofilm reactor[®] under conditions as described in Section 2 with the presence of chlorine (150 ppm). The graphs show the experimental data points (mean $\pm$ SD) and model curves of *L. plantarum* B1 (diamonds) and *L. innocua* FI2 (squares) determined by plate count method (open symbols and dotted line) and FISH (closed symbols and continued line). FISH, fluorescence in situ hybridization

safety risk. Regarding *L. plantarum*, Kubota et al. (2008) observed a substantial level of LAB isolate variability with respect to biofilm formation; however, they revealed that none of the isolates formed biofilm better than the *Lactobacillus plantarum* type strain. It was also reported that biofilm cells of *L. plantarum* M606 (onion isolate) were more resistant to ethanol and acetic acid than planktonic cells. Later studies revealed that biofilm cells showed greater resistance also to citric acid, lactic acid, malic acid, and sodium hypochlorite than their planktonic counterparts (Kubota et al., 2009). Similarly, sessile cells of *L. monocytogenes* appeared less sensitive to different biocides than the planktonic cells (Aarnisalo et al., 2000; Gram, Bagge-Ravn, Ng, Gymoese, & Vogel, 2007; Pan, Breidt, & Kathariou, 2006; Robbins, Fisher, Moltz, & Martin, 2005). Some studies have focused on the increased resistance of *L. monocytogenes* to disinfectants in mixed species biofilms, but often without comparing results with single species biofilm resistance, and as such it was impossible to conclude whether the increased resistance was dependent on the interaction

between species (Fatemi & Frank, 1999; Norwood & Gilmour, 2000). Others have shown that in mixed species biofilms of *L. monocytogenes* and *Salmonella enterica*, the interactions did not affect either the biofilm forming ability or the resistance of each species to several types of disinfectants (Kostaki, Chorianopoulos, Braxou, Nychas, & Giaouris, 2012). In turn, Giaouris, Chorianopoulos, Doulgeraki, and Nychas (2013) found an increased resistance of *Pseudomonas putida* to benzalkonium chloride (BAC) when grown in a mixed-species biofilm with *L. monocytogenes*, whereas biofilm mode of *Listeria* existence either mono or dual-species did not influence listeriae resistance. Opposite results were obtained by Saá Ibusquiza, Herrera, Vázquez-Sánchez, and Cabo (2012), revealing that the association of *L. monocytogenes* and *P. putida* accelerates biofilm formation and increases BAC-resistance of the biofilm compared to that formed by mono-species biofilms. At last, Van der Veen and Abee (2011) reported that mixed species biofilms of *L. monocytogenes* and *L. plantarum* showed enhanced resistance to BAC and peracetic acid. Although these findings provide a significant insight into biofilms

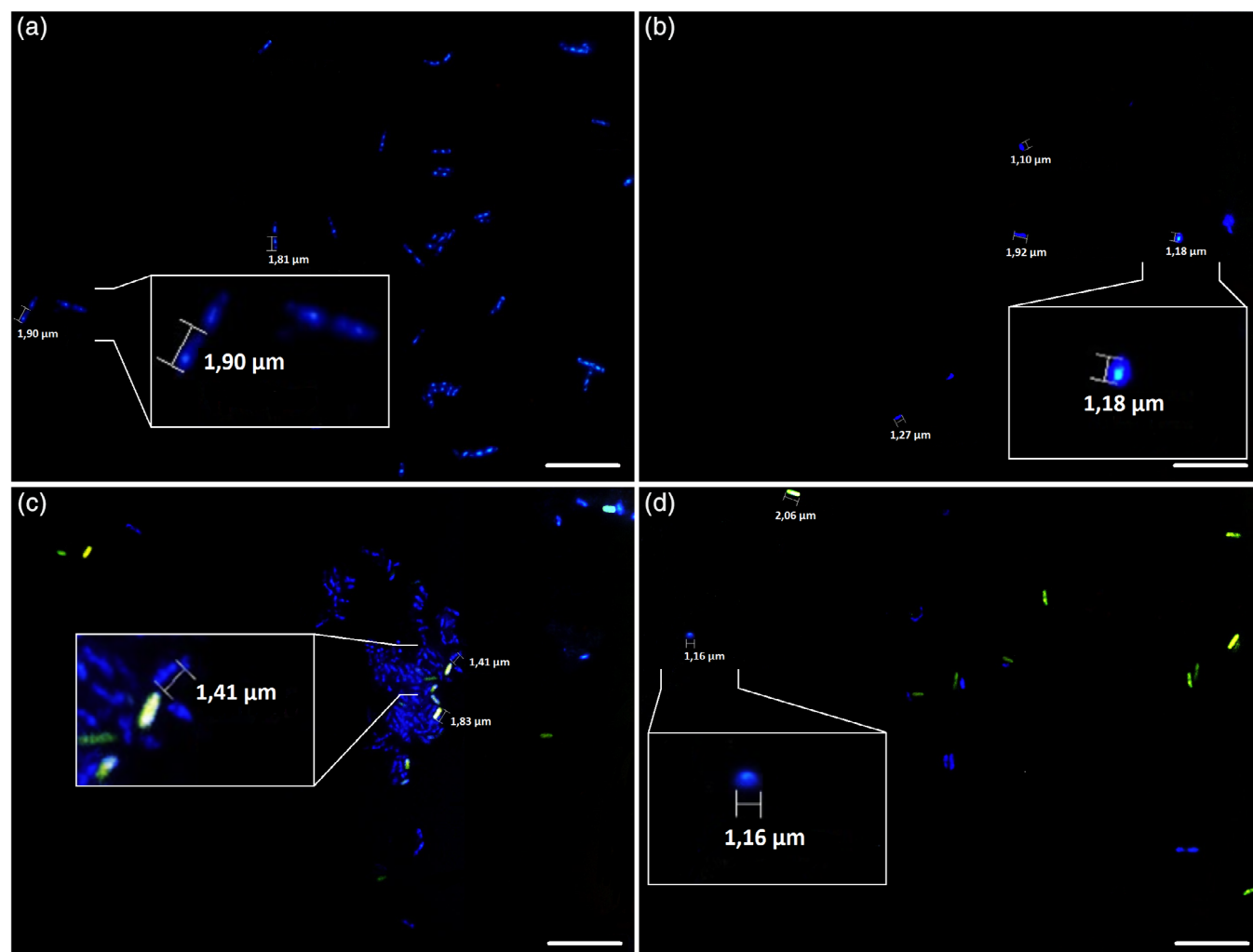


FIGURE 3 The epifluorescence microscopy (EFM) images of single and mixed species biofilms grown for 24 hr at 30°C. (a) Single species biofilm of *L. innocua* F12 cells fluoresce blue (no chlorine). (b) Single species biofilm of *L. innocua* F12 cells fluoresce blue (with chlorine). (c) Mixed species biofilm of *L. plantarum* B1 cells fluoresce yellow and *L. innocua* F12 cells fluoresce blue (no chlorine). (d) Mixed species biofilm of *L. plantarum* B1 cells fluoresce yellow and *L. innocua* F12 cells fluoresce blue (with chlorine). The scale bar is 10 μm

resistance to disinfectants, none of these studies considered the change in the physiological state of cells, as they relied largely on plate count results. In our study, we investigated single and mixed species biofilms of *L. plantarum* and *L. innocua* in dynamic conditions with and without chlorine by two methods of enumeration: plate counting for estimation of culturable cells and FISH for viable cells. We found a decrease in culturable cells and an increase in viable cells of these species, with the latter cells sustaining this status quo longer in a coculture. In the presence of chlorine, the early stage of growth resulted in the transient nonculturability of the cell populations and the nonculturable fraction was most substantial over the first 8-hr period in a coculture. As for *L. innocua*, almost complete nonculturability was observed, indicating that not only a chlorine challenge but also the interspecies interactions had the impact on the physiological state especially of listeriae cells.

Precise determination of the physiological state of cells and VBNC cells as a kind of viable cells requires a suitable methodological approach. Therefore, the first step in detecting VBNC cells involves the

estimation of the remaining culturable cells in the sample by a conventional plate count technique. The second major step includes the estimation of viable cells by detecting certain target sites within bacterial cells. One common approach involves the application of cell staining with a fluorescent marker and direct microscopic enumeration (Li et al., 2014). The rRNA content of bacterial cells is a good example here, because it can be quantified microscopically after fluorescence in situ rRNA hybridization (FISH). As rRNA content varies in bacteria depending on their growth rate and decreases fast enough in inactive cells, FISH was previously proposed as a suitable method to determine the physiological state of cells (Joux & Lebaron, 2000). Moreover, in complex communities, this assay could be developed to detect the activity of specific bacteria using appropriate nucleotide probes. In the latter case, we have adopted two species-specific probes to differentiate *L. plantarum* and *Listeria innocua* in a sample. However, one obstacle of applying FISH in this model was the permeability barrier posed by thick and highly anionic cell walls of these Gram-positive bacteria

(Brehm-Stecher et al., 2005). Despite the preparatory steps involving lysozyme digestion, the presence of two types of cells, brightly fluorescent and fading, especially in the case of *L. plantarum*, could not be avoided. This often happens when samples contain cells that differ markedly in their requirements for permeabilization, resulting in either under- or overdigestion, and thus in cell loss, and this was noticeably revealed for *L. plantarum* cells at the late stage of growth.

5 | CONCLUSION

It should be noted that a better understanding of the physiological state of cells, including the VBNC state, is necessary to determine the outgrowth potential of any microbe to ensure food quality and safety. However, this is especially important in the case of pathogenic bacteria, which may remain VBNC and retain virulence, for example, upon rapid resuscitation (Oliver, 2009). We showed that microbial interactions between *L. plantarum* and *L. innocua* induce changes in the physiological state of individual species, most probably adopting nonculturability when challenged with chlorine. We also found that cells of *L. innocua* change morphologically under stress, becoming much smaller, coccoid forms. Certainly, the morphological changes are commonly attributed to VBNC cells, however, similar changes also occur in non-VBNC cells under environmental stress conditions (Li et al., 2014), hence it seems appropriate in future research to provide the insight into the understanding of the molecular control of VBNC which may yield new information to improve detection and control methods for VBNC bacteria. It was previously assumed that in environmental conditions such as food processing environments, biofilms may consist of multiple species which occupy the same niche. Based on our findings, we consider our approach for future studies to test this hypothesis using isolates from specific sites of the food processing facility and estimate their performance in multi species experiments using a variety of conditions including surface materials, antimicrobial agents, different temperatures, and media composition.

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AUTHOR CONTRIBUTIONS

M.A.O. designed the study and performed the experiments. I.B. derived the model and performed the computations. M.A.O. and I.B. contributed to the interpretation of the results. M.A.O. wrote the manuscript.

ORCID

Magdalena A. Olszewska  <https://orcid.org/0000-0001-9980-9317>

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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